Automatic Recognition of Oil Spills Using Neural Networks and Classic Image Processing

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1 Introduction

Oil spills pose severe environmental, economic, and ecological threats to marine and coastal ecosystems. Rapid and accurate detection of oil spills is crucial to minimize long-term damage to marine life, fisheries, and coastal infrastructure.

Synthetic Aperture Radar (SAR) imagery is widely used for marine monitoring because it operates independently of daylight and weather conditions. Oil spills suppress short gravity-capillary waves on the sea surface, leading to reduced radar backscatter and dark regions in SAR images. However, similar dark patterns known as oil look-alikes—such as low wind areas, grease ice, and internal waves—make automatic detection challenging.

This paper addresses the challenge of distinguishing true oil spills from visually similar phenomena by combining convolutional neural networks (CNNs) with carefully selected classical image processing techniques.

2 Related Work and Motivation

Traditional oil spill detection approaches rely on thresholding, statistical analysis, and handcrafted texture features. These methods are sensitive to speckle noise and environmental variability.

Recent studies have demonstrated that CNN-based semantic segmentation significantly outperforms classical approaches. Architectures such as U-Net and DeepLabv3+ are widely used for pixel-level oil spill classification. However, CNNs alone struggle to distinguish oil spills from oil look-alikes due to subtle visual differences.

The key motivation of this paper is the hypothesis that emphasizing the physical and statistical characteristics of oil spills through image filtering can significantly improve CNN performance without losing critical information.

3 Dataset Description

The study uses the publicly available Oil Spill Detection Dataset introduced by Krestenitis et al. The dataset is derived from Sentinel-1 SAR imagery and includes annotations for five classes:

- Ocean
- Oil Spill
- Oil Look-Alike
- Ship
- Land

The dataset consists of 1002 images for training and 110 images for testing. Each image is accompanied by pixel-level ground truth masks.

4 CNN Architectures

The objective is semantic segmentation, where each pixel is classified into one of the five predefined classes. Two CNN architectures are evaluated:

4.1 U-Net

U-Net follows an encoder–decoder architecture with skip connections. The encoder extracts hierarchical features, while the decoder restores spatial resolution.

In this study, ResNet-101 is used as the encoder backbone due to its strong feature extraction capability and proven performance in prior oil spill studies.

4.2 DeepLabv3+

DeepLabv3+ improves segmentation performance by combining atrous spatial pyramid pooling (ASPP) with a decoder module. ASPP captures multiscale contextual information, while the decoder refines object boundaries.

MobileNetV2 is used as the encoder backbone to reduce computational complexity while maintaining accuracy.

5 Data Augmentation

The original dataset size is insufficient for robust CNN training. To address this, extensive data augmentation is applied during training.

Augmentation techniques include:

- Horizontal and vertical flipping
- Random rotation between $90°$ and $270°$
- Perspective transformations

Nearest-neighbor interpolation is used instead of bilinear interpolation to prevent incorrect label mixing at class boundaries. This ensures that segmentation masks remain physically consistent after transformation.

6 Image Filtering

To emphasize oil spill characteristics, classical image filtering is applied before CNN classification.

6.1 Histogram Equalization

Histogram equalization redistributes pixel intensities to produce a more uniform intensity distribution. This enhances contrast between oil spills and surrounding ocean regions, highlighting texture differences.

6.2 Contrast Stretching

Contrast stretching maps the lower and upper percentile intensity values to the full grayscale range:

$$P_{out} = (P_{in} - c) \cdot \frac{255}{d - c}$$

where c and d represent the 2nd and 98th percentile intensity values.

This filter enhances grayscale contrast while preserving local intensity relationships.

7 Model Ensemble

Individual CNN models exhibit noticeable generalization errors. To mitigate this, an ensemble approach is adopted.

Six trained models are combined:

- U-Net (no filter)
- DeepLabv3+ (no filter)
- U-Net with contrast stretching
- DeepLabv3+ with contrast stretching
- U-Net with histogram equalization
- DeepLabv3+ with histogram equalization

The final prediction is obtained by averaging the SoftMax probability outputs and selecting the class with the highest probability for each pixel.

8 Methodology

The complete oil spill detection pipeline consists of the following steps:

1. Preprocess Sentinel-1 SAR images.
2. Apply data augmentation to increase dataset diversity.
3. Enhance image characteristics using histogram equalization and contrast stretching.

4. Train U-Net and DeepLabv3+ architectures.

5. Combine model outputs using ensemble averaging.

6. Generate final pixel-level oil spill segmentation maps.

This hybrid framework integrates physical insight with data-driven learning.

9 Performance Analysis

Performance is evaluated using Intersection over Union (IoU) and mean IoU metrics:

$$IoU = \frac{|Prediction \cap GroundTruth|}{|Prediction \cup GroundTruth|}$$

The ensemble model achieves a mean IoU of 71.12%, representing a 9.3% improvement over the baseline DeepLabv3+ model. Oil spill IoU improves by approximately 4.2%, while oil look-alike classification improves by 6.2%.

Results show consistent improvement across most classes, confirming the effectiveness of combining image filtering with CNN-based segmentation.

10 Conclusion

This paper demonstrates that combining classical image processing techniques with CNN-based semantic segmentation significantly improves oil spill detection in SAR images.

Histogram equalization and contrast stretching enhance physically meaningful image characteristics, while ensemble learning reduces generalization error. The proposed framework achieves state-of-the-art performance and provides a practical solution for operational marine oil spill monitoring.

Future work includes refining oil look-alike categorization and extending the approach to additional SAR sensors.